Shrub Cover Analysis Report

Executive summary
The notebook extracts the SHR band from the 2025 RAP 10 m vegetation cover product and analyses a small south-west sample window from the selected California tile.
Shrub cover is widespread but mostly low: the mean is 12.07%, the median is 6%, and 95.85% of valid pixels are above zero.
The distribution is strongly right-skewed, indicating many sparse pixels and relatively few dense shrub patches.
The dominant class is 0–5% shrub cover (45.29% of valid pixels), while cover above 50% is rare (2.33%).
A 10% threshold is useful for a binary preview, but it simplifies low-cover shrub areas into an absence class.

This report explains the notebook workflow, summarises the key statistics, and interprets the six diagnostic figures produced by the analysis.

1. Study input and workflow

The uploaded notebook authenticates with Google Earth Engine, reads the RAP yearly vegetation cover collection, filters the collection to year 2025, and selects image ID 2025-10-500000-4300000. It then extracts the SHR band, renames it as shrub_cover, defines an approximately 5 km × 5 km rectangular region of interest in the south-west part of the tile, and converts that window to a NumPy array for statistical inspection.

The final notebook cell produces the following outputs:

- basic descriptive statistics and quantiles
- zero-cover versus positive-cover summary
- histogram and boxplot
- spatial image of the shrub-cover array
- cumulative distribution curve
- cover-class bar chart
- binary shrub presence preview using a 10% threshold

2. Dataset summary

Metric	Value
Image collection	RAP yearly vegetation cover (10 m)
Selected year	2025
Selected tile ID	2025-10-500000-4300000
Band analysed	SHR → shrub_cover

Metric	Value
Array shape	500 × 501
Total pixels	250,500
Valid pixels	231,500
NaN pixels	19,000
Approximate valid area	23.15 km^2

3. Statistical summary

Metric	Value
Mean shrub cover	12.07%
Median shrub cover	6%
Standard deviation	14.22%
Minimum	0%
25th percentile	2%
75th percentile	16%
Maximum	76%
Zero-cover pixels	9,598 (4.15%)
Positive-cover pixels	221,902 (95.85%)
90th / 95th / 99th percentiles	36% / 44% / 56%

These statistics show that shrubs are present across most of the valid area, but the typical pixel still has low cover. The median of 6% is much lower than the mean of 12.07%, which is a classic sign of a right-skewed distribution with a long upper tail.

4. Cover-class breakdown

Cover class (%)	Pixel count	Share of valid pixels (%)	Approx. area (ha)
0	9,598	4.15	95.98
0–5	104,842	45.29	1048.42
5–10	38,956	16.83	389.56
10–25	37,441	16.17	374.41
25–50	35,274	15.24	352.74
50–75	5,387	2.33	53.87

Cover class (%)	Pixel count	Share of valid pixels (%)	Approx. area (ha)
75–100	2	0.00	0.02

The 0–5% class dominates the sample and accounts for 45.29% of all valid pixels. When the 0%, 0–5%, and 5–10% classes are combined, about 66.27% of the valid area falls at or below 10% shrub cover. Conversely, cover above 50% is rare and occupies only 2.33% of the valid area.

5. Figure-based interpretation

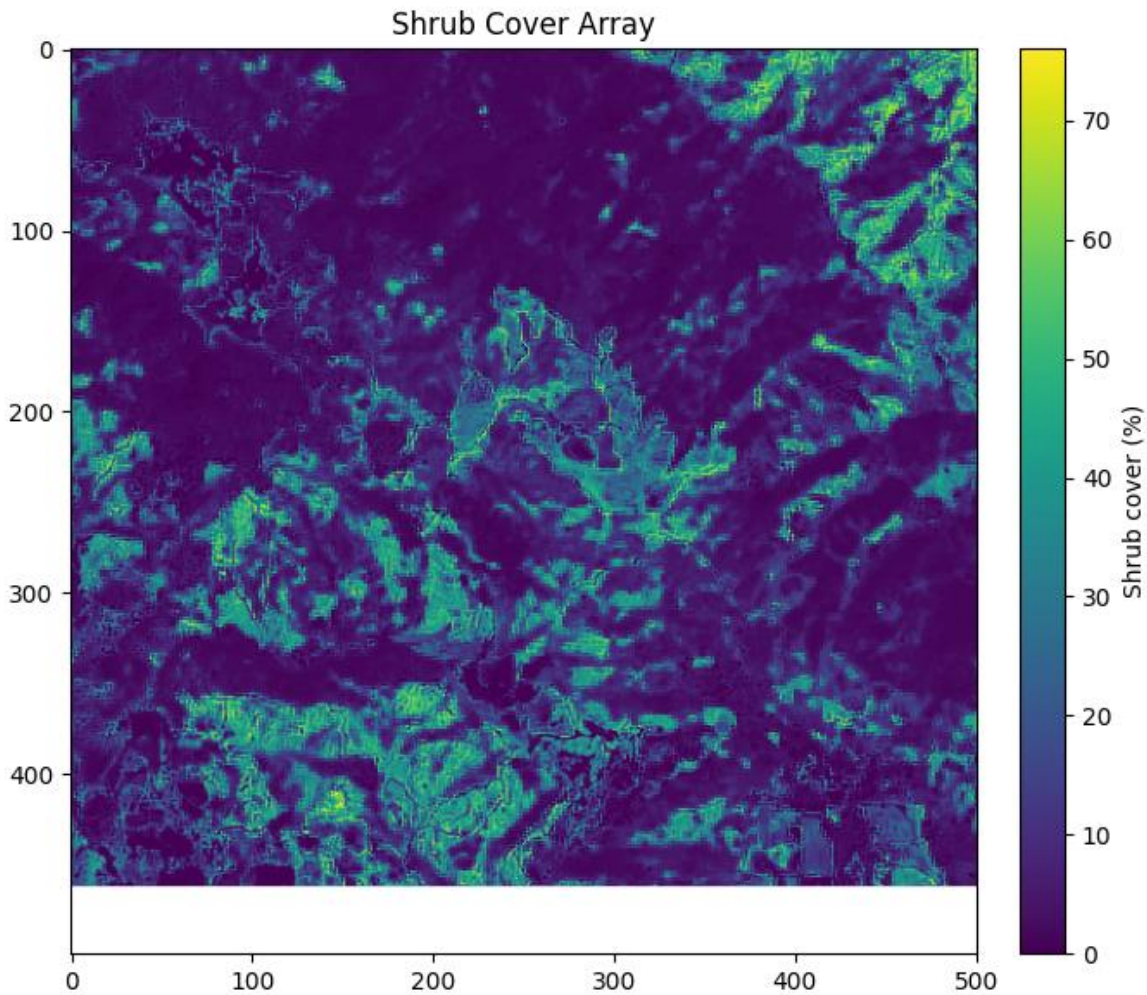

Figure 1. Spatial distribution of shrub cover values within the sampled window.

Interpretation: The spatial map shows a patchy and heterogeneous landscape. Higher shrub-cover zones occur in clusters rather than as one continuous dense shrub unit, which suggests local environmental controls and strong within-tile variability.

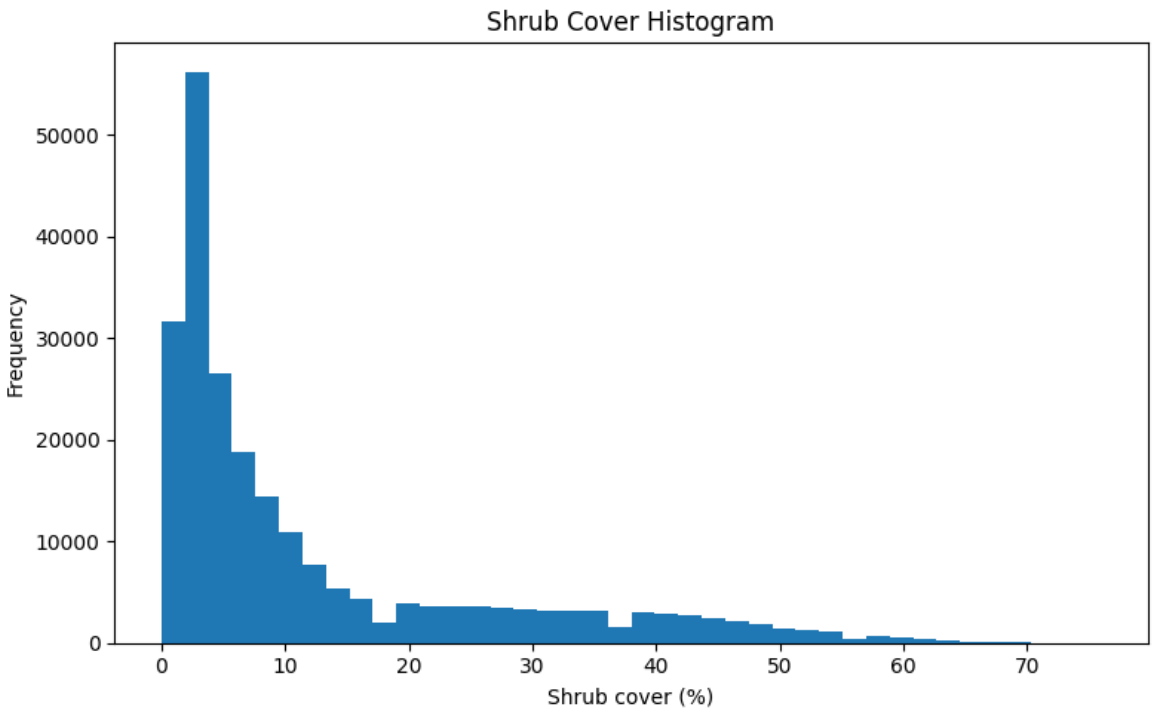

Figure 2. Histogram of shrub cover values.

Interpretation: The histogram is concentrated at low values and tails off gradually toward higher values. This confirms that sparse shrub cover is common, moderate cover is less common, and dense shrub cover is limited to a small subset of pixels.

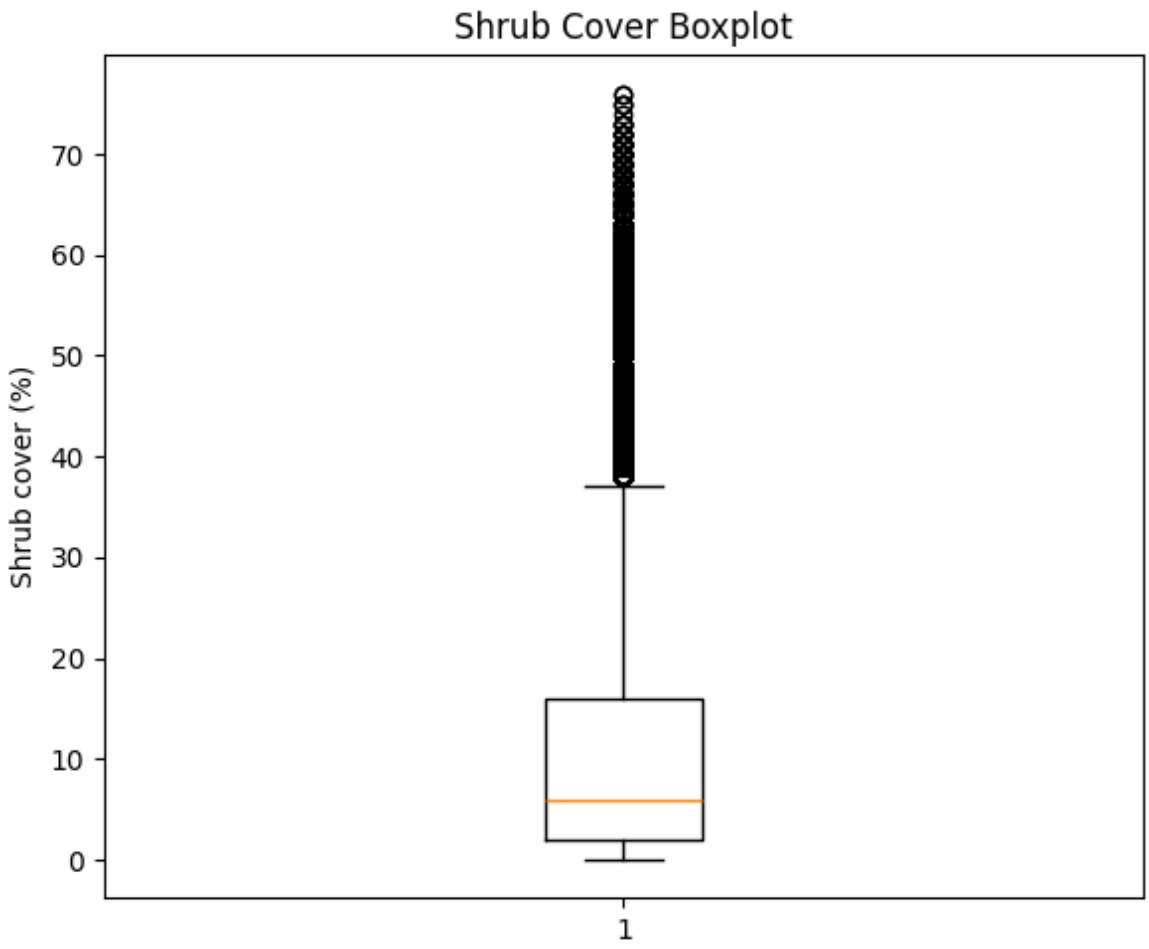

Figure 3. Boxplot of shrub cover values.

Interpretation: The boxplot highlights a low median and a wide upper tail with many high-value outliers. In practice, this means that high-cover shrub patches do exist, but they are limited and do not represent the dominant landscape condition.

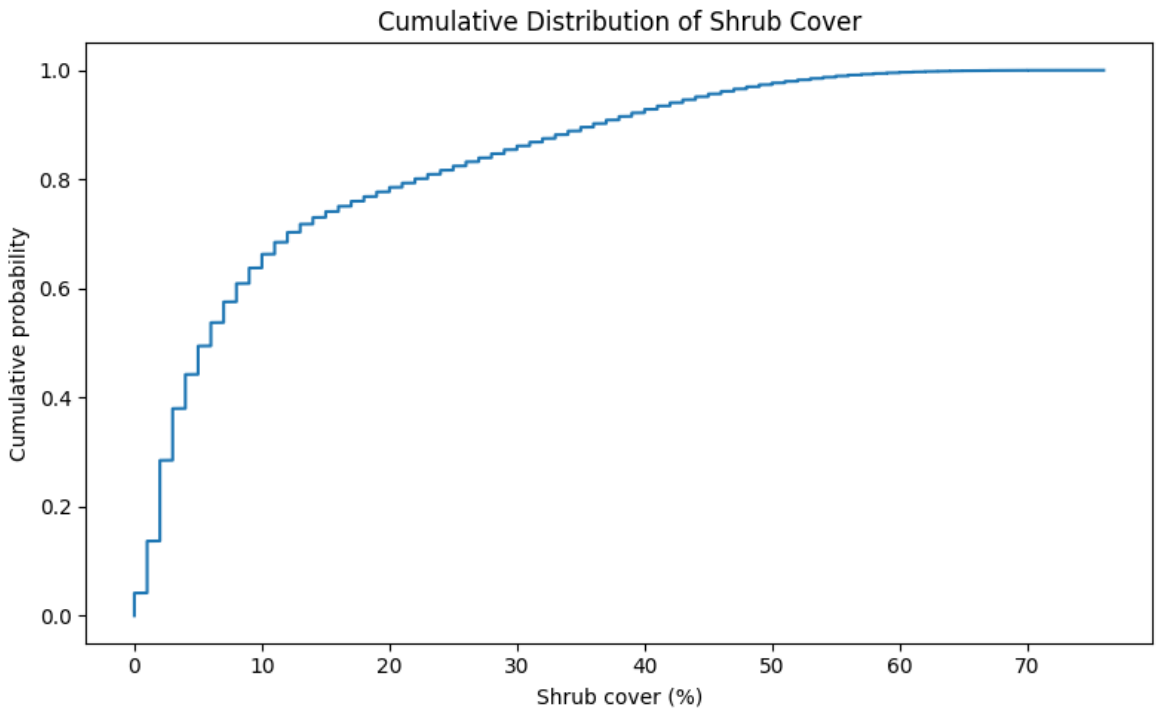

Figure 4. Cumulative distribution of shrub cover.

Interpretation: The cumulative curve rises quickly at low shrub-cover values, showing that a large fraction of the sample is accumulated before cover reaches moderate percentages. This makes the plot especially useful for threshold-based decisions such as choosing binary presence criteria.

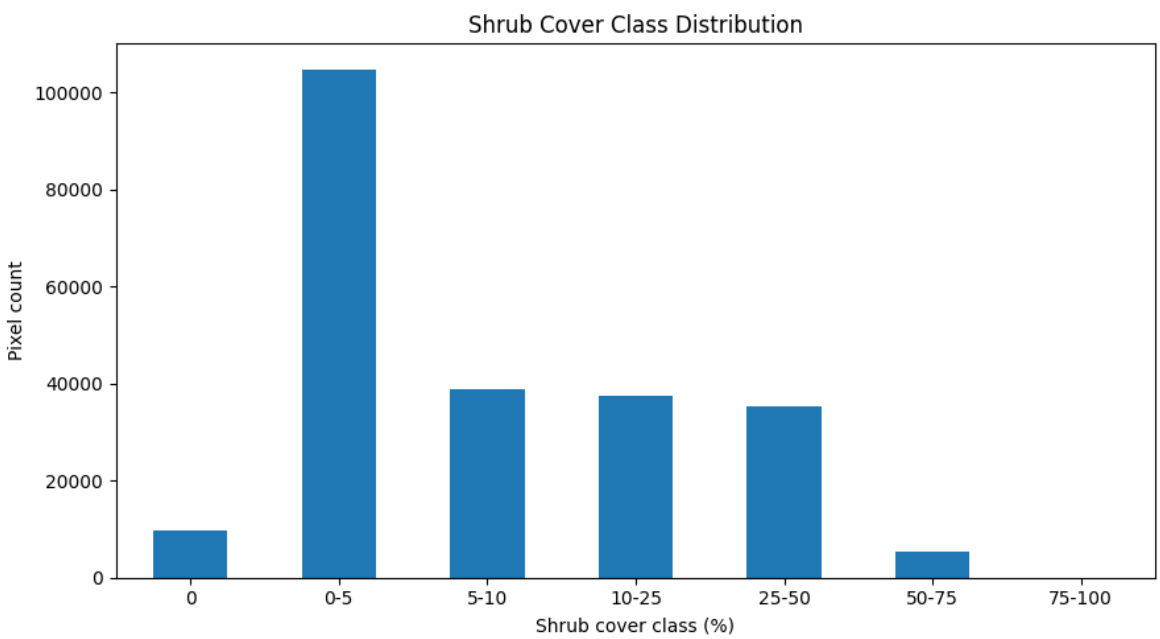

Figure 5. Shrub-cover class distribution.

Interpretation: The class chart gives the clearest practical summary: the 0–5% class is dominant, the 10–50% interval is important but secondary, and cover above 50% is uncommon. The sampled area is therefore best described as mostly low-cover shrubland with scattered moderate patches.

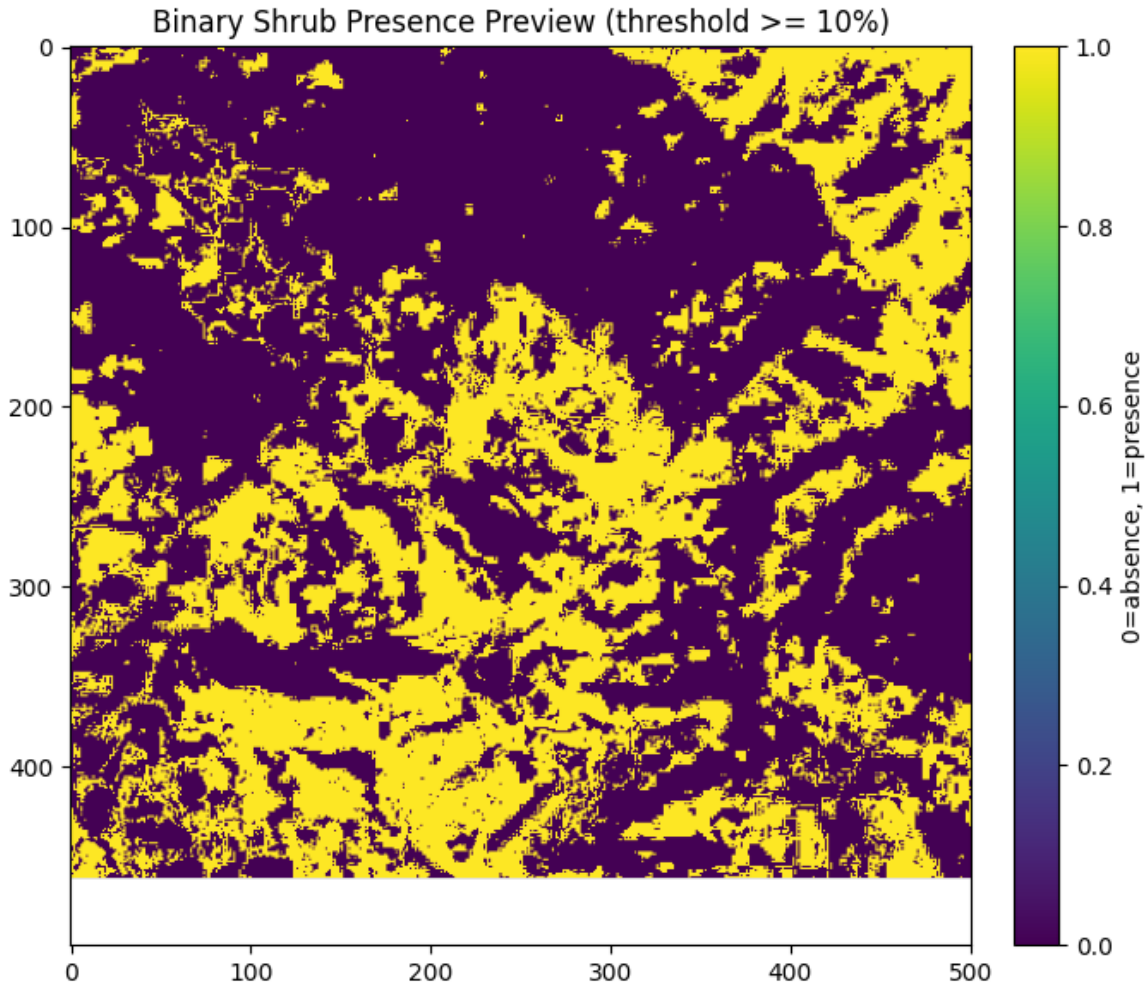

Figure 6. Binary shrub presence preview using a 10% threshold.

Interpretation: The binary preview is useful for rapid classification, screening, or downstream machine-learning tasks. However, it should be interpreted cautiously: pixels below 10% are treated as absence even though many of them still contain some shrubs. The threshold simplifies the landscape and is therefore analytical rather than ecological in a strict sense.

6. Main conclusions

- Shrub cover is present in most valid pixels, but predominantly at low percentages.
- The value distribution is strongly right-skewed, with a long upper tail and relatively few dense-shrub pixels.
- The area is spatially heterogeneous and contains localised clusters of moderate to high shrub cover.
- The 10% threshold is acceptable for a quick binary preview, but it should not be mistaken for a complete ecological definition of shrub absence.
- This notebook is a strong exploratory first step and can be extended to compare multiple tiles, years, or threshold settings.

Shrub Challenge Module Report

Interpretation of classification plots, strengths, weaknesses, and recommendations

Prepared based on the model plots shared by Sefa Karabas and Berkay Ersus

Key metrics

Metric	Value	Interpretation
Accuracy	0.649	Moderate overall performance; not sufficient alone because the dataset is highly imbalanced.
Balanced Accuracy	0.741	Encouraging result; suggests the model is not relying only on the dominant class.
Macro Precision	0.352	Low; indicates many false positives, especially for rare classes.
Macro Recall	0.741	High; the model tends to find minority classes rather than miss them.
Macro F1	0.342	Weak class-balanced quality; precision and recall are not well balanced across classes.
Weighted F1	0.724	Reasonable performance on majority classes, but this hides weakness on rare classes.
Cohen's Kappa	0.295	Low-to-moderate agreement beyond chance; the model contains signal but is not yet strong.

Executive summary

The current module shows stable but imbalanced classification behavior. It is clearly learning meaningful structure from the data, yet its performance is uneven across classes.

The strongest signs are the relatively good balanced accuracy, stable cross-validation results, and physically meaningful feature importance values in which z and intensity dominate the model.

The main limitation is the very low precision for rare classes, especially classes 7 and 18. In practice, this means the model can detect minority classes, but it does so with many false alarms.

As a result, the module is better suited for high-recall candidate screening than for final precise class assignment.

Detailed interpretation

The class support plot shows severe imbalance: class 1 has 337,307 samples, class 2 has 61,695, class 7 has 883, and class 18 has only 115. This imbalance strongly affects learning difficulty and metric interpretation.

Class 1 is the cleanest predicted class. Its precision is very high, meaning that when the model predicts class 1 it is usually correct. However, recall is notably lower, so many true class 1 samples are still being reassigned to other classes.

Class 2 has acceptable recall but weak precision. The model can find much of this class, but it also attracts many points from other classes, which lowers trust in individual class 2 predictions.

Class 7 is the weakest operational class. Although its recall is not low, its precision is near zero. This means the model labels many points as class 7 that are not actually class 7.

Class 18 is unusual: recall is very high even though the class is extremely small, but precision remains very low. This suggests an aggressive strategy that catches true class 18 points while also creating many false positives.

The confusion matrix confirms the main error pattern: many true class 1 points are reassigned to classes 2 and 7. This is the dominant source of confusion in the current module.

Feature importance is a positive sign. The top two predictors are z (0.506) and intensity (0.215), while x and y are lower. This suggests the model is using physically relevant information rather than only memorizing location, although some spatial dependence may still remain.

Prediction confidence is high for many samples, especially near 0.9 to 1.0. However, because precision is weak for some classes, the model is likely overconfident on part of its incorrect predictions.

Strengths

- Balanced accuracy is relatively good, indicating the model is not driven only by the majority class.
- Cross-validation mean +- std is very stable, which suggests the behavior is reproducible across folds.
- Feature importance ranking is physically sensible: elevation and intensity dominate.
- Minority classes are not completely missed; class 18 in particular has very high recall.
- Class 1 predictions are highly reliable when the model assigns that label.

Weaknesses

- Macro F1 is low, showing weak class-balanced prediction quality.
- Precision is very poor for rare classes, especially classes 7 and 18.
- The confusion matrix shows major leakage from class 1 into classes 2 and 7.
- The dataset is severely imbalanced, making some metrics unstable for minority classes.
- High confidence does not always mean correctness; some false positives appear to be predicted with strong confidence.

Recommendations

- Treat this module as a candidate-screening component rather than a final decision module.
- Tune class-specific decision thresholds using predicted probabilities.
- Consider stronger imbalance handling, including class weights, focal loss alternatives, or more constrained resampling.
- Add neighborhood-based geometric features such as local roughness, height variation, density, and return structure.
- Test a model variant without x and y to measure possible spatial leakage.
- Use post-processing or calibration for rare classes before deployment.

Overall conclusion

This module is promising as an early detection layer because it is sensitive to minority classes and stable under cross-validation. However, it is not yet selective enough for final operational classification. Its current profile is best described as high recall with low precision for rare classes.

In short, the model can help identify potential shrub-related candidate points, but it still needs refinement before it can support precise, low-false-alarm decisions.